

TRUNG V. HA
Research Assistant, University of Massachusetts Amherst

Email: vinhtrungha99@gmail.com | ORCID: [0000-0001-6600-2517](https://orcid.org/0000-0001-6600-2517)

GitHub: [tvh0021](https://github.com/tvh0021) | LinkedIn: [trung-ha-24a16v](https://www.linkedin.com/in/trung-ha-24a16v)

Curriculum Vitae

(Last updated: May 3, 2026)

TECHNICAL SKILLS

Programming	Python, C++, MATLAB, SQL
Packages	NumPy, Pandas, matplotlib, scikit-learn, SciPy, Astropy, JAX
Testing & Deployment	pytest, Sphinx, ReadtheDocs, GitHub Actions
Tools	Linux, Git, Jupyter, VSCode, Slurm/PBS, MPI
ML & Statistics	self-organizing maps, clustering, ensemble methods, hypothesis testing, cross-validation, Monte Carlo resampling, Fourier analysis

SELECTED PROJECTS

Anomaly Detection in 3D Datasets with Machine Learning

- Developed and shipped an open-source Python library, *aweSOM*, that detects turbulence features using self-organizing maps and statistical ensembles, trained on high-resolution simulation snapshots.
- Engineered convolutional features to highlight distinct structures.
- Implemented JAX for GPU acceleration; achieved up to 100x speedup over CPU baseline, enabling near-real-time inference at scale.
- Designed scaling and unit tests with pytest, wrote detailed documentation for external adoption with Sphinx.

Statistical Analysis of Stellar Groups Motion

- Built Pandas ETL pipelines to clean, transform, and validate data quality across millions of observations.
- Processed Gaia astrometric catalogs in FITS format, leveraging Astropy utilities for coordinate transformations and unit handling.
- Implemented resampling methods (Monte Carlo, jackknife, bootstrap) to quantify systematic uncertainties from noisy data.
- Developed a parallelized two-point correlation function, allowing 20x more samples in each calculation.

WORK & RESEARCH EXPERIENCE

Sep 2024 – present	Graduate Research Assistant , University of Massachusetts Amherst Design and run large-scale computational experiments modeling plasma physics around supermassive black holes with C++ & MPI. Analyze turbulent time-series signals using Fourier decomposition and power spectrum analysis to identify characteristic frequencies and scaling behavior in nonlinear dynamical systems. Secured NSF supercomputing allocation as PI; manage project timelines and computational resources.
Jun 2021 – Aug 2024	Graduate Research Assistant , University of North Texas Led end-to-end analysis for three first-author research projects, including a 22-author collaboration: synthesized ideas between experts into a unified analysis and peer-reviewed publication. Migrated a legacy MATLAB pipeline to Python and established a reproducible workflow adopted as the standard by subsequent researchers.
Sep 2023 – May 2024	Research Analyst (Pre-doctoral Fellow) and Guest Researcher , Center for Computational Astrophysics, Simons Foundation Developed open-source analysis software for plasma astrophysics as part of a cross-disciplinary collaboration between CCA researchers. Advanced scientific knowledge and numerical methodologies, communicated progress with both technical and non-technical audiences through 5+ public talks, and published results in two first-author papers.
Aug 2020 – May 2021	Graduate Teaching Assistant , University of North Texas
Sep 2018 – May 2020	Undergraduate Research Student , Center for Computational Relativity and Gravitation, Rochester Institute of Technology
Sep 2018 – Dec 2019	Undergraduate Teaching Assistant , University of Rochester
Jun 2018 – Aug 2018	Summer Research Intern , Laboratory for Laser Energetics, University of Rochester

EDUCATION

2024 – present	University of Massachusetts Amherst, Amherst, Massachusetts PhD candidate in Astronomy, expected graduation August 2026 Thesis title: “ <i>The Next Generation of Diagnostic Tools for Astrophysical Turbulence</i> ” PhD advisor: Yuan Li GPA: 4.00 / 4.00
2020 – 2024	University of North Texas, Denton, Texas PhD candidate – transferred to UMass before receiving degree Master of Science in Physics – conferred May 2022 GPA: 4.00 / 4.00 Relevant coursework: Numerical Methods, Machine Learning, Computer Vision

2017 – 2020

University of Rochester, Rochester, New York

Bachelor of Science in Physics

Relevant coursework: Applied Statistics, Computational Physics

SELECTED TALKS

- Dec 2025 **Stars & Plasma Astrophysics Group Meeting**, Flatiron Institute, New York, NY, USA
Title: “Modeling Multiphase Accretion Flows onto M87* with Galaxy- and Meso-scale Simulations”
- Jun 2025 **INVITED - Vietnam National Space Center**, Astrophysics Department Seminar, Hanoi, Vietnam
Title: “Feedback-driven Multiphase Accretion in M87: Results from Meso-scale Simulations”
- Jun 2025 **IAUS 397 – UniversAI: Exploring the Universe with Artificial Intelligence**, Athens, Greece
Title: “aweSOM: an Open-source Python Package for Efficient Clustering of Intermittency in Magnetized Plasma Turbulence”
- Mar 2025 **Stars & Plasma Astrophysics Group Meeting**, Flatiron Institute, New York, NY, USA
Title: “Segmentation of Single and Double Current Sheets in Magnetized Plasma Turbulence with Machine Learning”
- Jun 2024 **AstroAI Workshop**, Harvard-Smithsonian Center for Astrophysics, Cambridge, MA, USA, and
Midwest Magnetic Fields Workshop, Madison, WI, USA
Title: “Segmentation of Current Sheets in Magnetized Plasma Turbulence with Computer Vision”
- Apr 2024 **INVITED - Center for Computational Relativity and Gravitation Lunch Talk**, Rochester, NY, USA
Title: “Can Neural Networks Recognize Current Sheets? Using Computer Vision to Analyze Magnetized Plasma Turbulence”
- Feb 2024 **Kavli Institute for Theoretical Physics (KITP) – Turbulence in the Universe Workshop**, Santa Barbara, CA, USA
Title: “Segmentation of Current Sheets in Magnetized Plasma Turbulence with Computer Vision”
- Dec 2023 **Black Holes on Broadway: The Next Generation of AGN Models in Galaxy Formation**, New York, NY, USA
Title: “Bridging the Gap: Modeling Supermassive Black Holes Feeding and Feedback at the Meso-scale”
- Aug 2022 **Star Formation in Different Environments 2022**, Rencontres du Vietnam, Quy Nhon, Vietnam
Title: “Turbulence in Milky Way Star-forming Regions Traced by Young Stars and Gas”

PROFESSIONAL SERVICE

Served as referee for the Astrophysical Journal Letters.

Mentored two undergraduate students (Alexis NieMeira, Jonathan Red), one post-baccalaureate fellow (Benjamin Velguth), and two junior graduate students (Jacob Reinheimer, Roy Hsu), resulting in two publications (Velguth et al. 2025 and Reinheimer et al. 2026, as third author).

SUPERCOMPUTING & OBSERVING AWARDS

Dec 2024	Co-I., XMM-Newton proposal Title: “Identifying a Robust and Practical Accretion-Rate Indicator for Distant Quasars”; observing time: 110 kiloseconds.
Mar 2024	P.I., NSF ACCESS Explore allocation Title: “Turbulent multiphase accretion flows from supermassive black hole feedback”; amount: 400,000 ACCESS credits

AWARDS AND HONORS

Spring 2025	Mary Dailey Irvine Graduate Travel Grant, UMass Amherst, \$1000
2023 – 2024	The Zhibing Hu Scholarship, University of North Texas, \$1000.
May 2023	Featured on University of North Texas College of Science news: “UNT Physics Graduate Student Selected for Prestigious Pre-Doctoral Program” (link to article)
Spring 2023	College of Science Travel Award, University of North Texas. \$500.
Feb 2022	Featured on University of North Texas’s Graduate Research Spotlight .
2021 – 2025	R. B. Toulouse Scholarship, University of North Texas. \$1000/year.
Apr 2021	Featured on the <i>North Texan</i> : “Turbulent Motion Moves Research Forward” (link to article).
2019 – 2020	Take Five Scholar, University of Rochester. Thesis: “Exploring the Advantages and Shortcomings of French Literature in Translation”.

OTHER ACTIVITIES

Participated in the Flatiron Institute’s Center for Computational Astrophysics Pre-doctoral program in New York City in fall 2023.

Organizer for the weekly joint-UNT/UTD astronomy journal club, 2023.

Interviewed on AAS Journal Author Series about publication titled: “Measuring Turbulence with Young Stars in the Orion Complex” on [YouTube](#), 2021.

Participated in student exchange programs: “Cultural Exchange Program” in Arizona, USA during the 2014-15 school year and “French in France” in Rennes, France in summer 2019.

Citizenship: Vietnam

U.S. Permanent Resident

TRUNG V. HA
Research Assistant, University of Massachusetts Amherst

Email: vinhtrungha99@gmail.com | ORCID: [0000-0001-6600-2517](https://orcid.org/0000-0001-6600-2517)
GitHub: [tvh0021](https://github.com/tvh0021) | LinkedIn: [trung-ha-24a16v](https://www.linkedin.com/in/trung-ha-24a16v)

Publication List

In reverse chronological order, refereed publications are listed first

FIRST AUTHOR PUBLICATIONS

1. ***“Machine-Learning Characterization of Intermittency in Relativistic Pair Plasma Turbulence: Single and Double Sheet Structures” ****
Ha, Trung; Nästtilä, J.; Davelaar, J.; Sironi, L. ([2025ApJ...985L..31H](#))
2. ***“aweSOM: a CPU/GPU-accelerated Self-organizing Map and Statistically Combined Ensemble Framework for Machine-learning Clustering Analysis” ****
Ha, Trung; Nästtilä, J.; Davelaar, J. ([2025.JOSS.07613](#), [GitHub](#))
3. ***“Shedding New Light on Weak Emission-Line Quasars in the CIV–H β Parameter Space”***
Ha, Trung; Dix, C.; Matthews, B. M.; Shemmer, O.; et al., ([2023ApJ...950...97H](#))
4. ***“Turbulence in Milky Way Star-forming Regions Traced by Young Stars and Gas”***
Ha, Trung; Li, Y.; Kounkel, M.; Xu, S.; Li, H.; Zheng, Y., ([2022ApJ...934....7H](#))
5. ***“Measuring Turbulence with Young Stars in the Orion Complex”***
Ha, Trung; Li, Y.; Xu, S.; Kounkel, M.; Li, H., ([2021ApJ...907L..40H](#))
6. ***“Multi-phase Accretion Flows onto M87: Galaxy- and Meso-scale Simulations” ****
Ha, Trung; Li, Y., (in prep.)

OTHER PUBLICATIONS

7. ***“Probing Turbulence, Gravity, Supernovae, and Magnetic Field Effects with the 6D Kinematics of Young Stars in Milky Way Star-forming Regions”***
Velguth, B.; Li, Y.; Ha, Trung, et al. ([2025ApJ...990..165V](#))
8. ***“Rest-Frame Optical Spectroscopy of Ten $z \sim 2$ Weak Emission-Line Quasars”***
Chen, Y.; ...; Ha, Trung, et al. ([2024ApJ...972..191C](#))
9. ***“Gemini Near Infrared Spectrograph - Distant Quasar Survey: Rest-Frame Ultraviolet-Optical Spectral Properties of Broad Absorption Line Quasars”***
Ahmed, H.; ...; Ha, Trung, et al., ([2024ApJ...968...77A](#))
10. ***“The Nature of the Motions of Multiphase Filaments in the Centers of Galaxy Clusters”***
Ganguly, S.; ...; Ha, Trung, ([2023FrASS..1038613](#))
11. ***“Handing-Off the Outcome of Binary Neutron Star Mergers for Accurate and Long-Term Post-Merger Simulations”***
Lopez Armengol, F. G.; ...; Ha, Trung; et al., ([2022PhRvD.106h3015L](#))
12. ***“HARM3D+NUC: A new method for simulating the post-merger phase of binary neutron star mergers with GRMHD, tabulated EOS and neutrino leakage”***
Murguia-Berthier, A.; ...; Ha, Trung, et al., ([2021ApJ...919...95M](#))
13. ***“Black Hole-Host Galaxy Correlations with Machine Learning: A Comparative Study of Illustris, TNG, and EAGLE”***
Reinheimer, J.; Li, Y.; Ha, Trung, et al. (submitted to OJA, [arXiv:2510.20751](#))